

*MA.7.AR.3.3***Benchmark**

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Benchmark Clarifications:

Clarification 1: Problem types are limited to length, area, weight, mass, volume and money.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.2
- MA.7.AR.2.2
- MA.7.AR.4.3, MA.7.AR.4.4, MA.7.AR.4.5
- MA.8.GR.1.2

Terms from the K-12 Glossary

- Area
- Capacity
- Customary Units
- Metric Units

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.3.5
- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3

Next Benchmarks

- MA.912.FL.2.4

Purpose and Instructional Strategies

In previous courses, students performed conversions within the same measurement system. In grade 7 accelerated, students solve mathematical and real-world problems involving the conversion of units across different measurement systems. Students will also apply these conversions when solving problems involving the distance between two points in a coordinate plane. In future courses, students will convert between currencies given exchange rates.

- Focus on using conversion ratios to create equivalent values.
 - For example, if 1 foot = 12 inches, you can use the ratio of $\frac{1}{12}$ to solve problems. Students may also use conversion ratios in their science courses. Emphasize that multiplying by equivalent values of 1 does not change the value but gives an equivalent value in another unit of measurement.
- Instruction includes using manipulatives to estimate conversion ratios across measurement systems such as yardsticks, meter sticks, measuring cups and graduated cylinders (*MTR.2.1*).
- Students may need review on which units are used to measure length, volume and mass.
- Instruction includes using a reference sheet with conversion ratios.

Common Misconceptions or Errors

- Students may incorrectly place the values in a conversion ratio. To address this misconception, have students estimate values prior to calculations using the conversion ratio (*MTR.6.1*).

Strategies to Support Tiered Instruction

- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - What do you know from the problem?
 - What is the problem asking you to find?
 - Can you create a visual model to help you understand the problem?
- Teacher provides unit conversion sheet for students to determine the unit of measurement between different systems.
- Instruction focuses on using a unit conversion table to explicitly describe the process of converting between different units of measurement.
- Teacher provides instruction on color-coding and labeling the different units when setting up a proportional relationship to ensure corresponding units are placed in the corresponding positions within the proportion.
- Teacher encourages students to use their prior knowledge of proportions to convert units of measurement across different measurement systems.

$$\frac{30m}{ft} = \frac{9.5ft}{x}$$

x 9.5

- Teacher provides students a visual of units of measurement between different systems to compare and estimate the proper conversion ratio.
- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Teacher has students estimate values prior to calculations using the conversion ratio (*MTR.6.1*).

Instructional Tasks

Instructional Task 1 (MTR.1.1, MTR.4.1)

Joe was planning a business trip to Canada, so he went to the bank to exchange \$200 U.S. dollars for Canadian (CDN) dollars. On the way home from the bank, Joe's boss called to say that the destination of the trip had changed to Mexico City. Joe went back to the bank to exchange his Canadian dollars for Mexican pesos. What is the value of Mexican pesos that Joe has now?

Part A. What questions still need to be answered to approach this problem?

Part B. The rates for CDN to the U.S. dollar and the rate of pesos to the CDN are shown below.

- Rate of \$1.02 CDN per \$1 U.S.
- Rate of 20.8 pesos per \$1 CDN

What is the value of Mexican pesos that Joe has now?

Instructional Items

Instructional Item 1

Mary is buying strings of lights to hang on her patio deck. She needs 80 feet of lights to go around the entire patio, but the lights she wants to buy are only sold in packs of 5 meters. If one meter is approximately 3.28 feet, how many packs of lights will Jia need for her patio?

Instructional Item 2

How many milliliters are in 12 fluid ounces?

Instructional Item 3

Convert 50 pounds to kilograms.

Instructional Item 4

When driving from Germany to Poland, the speed limit signs change from miles per hour (mph) to kilometers per hour (kph), but your rental car speedometer only reads in mph. If the speed limit on the highway is 100 kph, at what speed will you exceed the speed limit?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*